H4: Trace Methods

Following Griffiths' Exact Procedure

From Example 7.5/7.6 through Eq. 7.129 to the Møller Cross Section

Every step shown as Griffiths does it — Casimir's trick, trace evaluation, index contraction

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§1 Griffiths' Method: The Overview

How Griffiths Does It (Sections 7.7–7.8)

Griffiths' procedure has very specific steps. Here they are in order, exactly as they appear in the textbook:
Step A (Example 7.5, Eq. 7.126): Apply **Casimir's trick** to the squared amplitude. This converts spin sums into a *product of two traces* — one for each fermion line.
Step B (Example 7.6, Eq. 7.128): Evaluate each trace *individually* using **trace theorem 13** (the four-gamma-matrix trace formula). Each trace produces three terms involving momenta with free μ, ν indices.
Step C (Eq. 7.129): **Multiply** the two trace results together and **contract** the μ, ν indices. This produces $3 \times 3 = 9$ terms (or 12 with the mass terms). Most simplify or cancel, leaving the final answer.
We'll walk through each step in painful detail.

§2 Step A: Casimir's Trick (Griffiths Eq. 7.126)

§2.1 Starting Point: The Squared Amplitude

For the direct term $|\mathcal{M}_1|^2$ in Møller scattering (which has the same structure as $e\mu$ scattering):

$$|\mathcal{M}_1|^2 = \frac{g_e^4}{(p_1 - p_3)^4} \, [\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)] \, [\bar{u}(3)\gamma^\nu u(1)]^*[\bar{u}(4)\gamma_\nu u(2)]^*$$

§2.2 What Casimir's Trick Does

The Key Move: Spin Sums Factor

When you sum over all spins, the electron current and "muon" current **decouple** into separate sums:

$$\frac{1}{4}\sum_{\text{spins}} |\mathcal{M}_1|^2 = \frac{g_e^4}{4(p_1 - p_3)^4} \underbrace{\text{Tr}[\gamma^\mu(\not{p}_1 + m)\gamma^\nu(\not{p}_3 + m)]}_{\text{“electron trace”}} \times \underbrace{\text{Tr}[\gamma_\mu(\not{p}_2 + M)\gamma_\nu(\not{p}_4 + M)]}_{\text{“muon trace”}}$$

Why two separate traces? Because in $|\mathcal{M}_1|^2$, the electron bilinears pair up with each other ($\bar{u}(3)\ldots u(1)$ with $[\bar{u}(3)\ldots u(1)]^*$), and the muon bilinears pair up with each other. The spin sums are independent. Each pair → one trace.
Setting $m = M = 0$ for our massless problem:

$$\langle|\mathcal{M}_1|^2\rangle = \frac{g_e^4}{4(p_1 - p_3)^4} \, \text{Tr}[\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_3] \times \text{Tr}[\gamma_\mu \not{p}_2 \gamma_\nu \not{p}_4]$$

Reading the Trace from the Diagram

Griffiths tells you to follow each fermion line **backwards** (against the arrow). For the electron line $p_1 \to p_3$:

Start at the adjoint spinor $\bar{u}(3)$, then the vertex γ^μ, then the spinor $u(1)$. From the conjugate: $\bar{u}(1)$, then γ^ν, then $u(3)$. The completeness relation turns $u(1)\bar{u}(1) \to \not{p}_1$ and $u(3)\bar{u}(3) \to \not{p}_3$. The trace closes the loop.
Result: $\mathrm{Tr}[\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_3]$.
The $\not{p}$ associated with the adjoint spinor (the one with the bar) appears *after* the gamma matrix on its left. Griffiths notes this ordering rule explicitly.

§3 Step B: Evaluating Each Trace (Griffiths Example 7.6)

This is the heart of Griffiths' method. We evaluate $\mathrm{Tr}[\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_3]$ using **trace theorem 13**.

§3.1 The Tool: Trace Theorem 13

Trace Theorem 13 — The Four-Gamma Trace

$$\mathrm{Tr}(\gamma^\alpha \gamma^\beta \gamma^\gamma \gamma^\delta) = 4(g^{\alpha\beta} g^{\gamma\delta} - g^{\alpha\gamma} g^{\beta\delta} + g^{\alpha\delta} g^{\beta\gamma})$$

Pattern: **first–second** × **third–fourth**, minus **first–third** × **second–fourth**, plus **first–fourth** × **second–third**.
Think of it as: pair the four indices in all possible ways that preserve the cyclic order, with alternating signs $+, -, +$.

§3.2 Applying Theorem 13 to the Electron Trace

Write $\not{p}_1 = \gamma^\lambda (p_1)_\lambda$ and $\not{p}_3 = \gamma^\sigma (p_3)_\sigma$. Then:

$$\mathrm{Tr}[\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_3] = (p_1)_\lambda \, (p_3)_\sigma \; \mathrm{Tr}[\gamma^\mu \gamma^\lambda \gamma^\nu \gamma^\sigma]$$

Pulling Out the Momenta

The momenta $(p_1)_\lambda$ and $(p_3)_\sigma$ are just *numbers* (components of four-vectors), not matrices. They commute with everything and can be pulled out of the trace. What's left inside the trace is four bare gamma matrices: γ^μ, γ^λ, γ^ν, γ^σ.

Apply Theorem 13 with $\alpha = \mu,\ \beta = \lambda,\ \gamma = \nu,\ \delta = \sigma$:

$$\mathrm{Tr}[\gamma^\mu \gamma^\lambda \gamma^\nu \gamma^\sigma] = 4(g^{\mu\lambda} g^{\nu\sigma} - g^{\mu\nu} g^{\lambda\sigma} + g^{\mu\sigma} g^{\nu\lambda})$$

Now contract with $(p_1)_\lambda \, (p_3)_\sigma$. Each metric tensor g contracts with a momentum to raise/lower an index:

$$\begin{aligned} g^{\mu\lambda}(p_1)_\lambda &= p_1^\mu & g^{\nu\sigma}(p_3)_\sigma &= p_3^\nu \\ g^{\lambda\sigma}(p_1)_\lambda (p_3)_\sigma &= p_1 \cdot p_3 & & \\ g^{\mu\sigma}(p_3)_\sigma &= p_3^\mu & g^{\nu\lambda}(p_1)_\lambda &= p_1^\nu \end{aligned} \tag{1}$$

Putting it together:

Electron Trace — Griffiths Eq. 7.128 (with $m = 0$)

$$\boxed{\mathrm{Tr}[\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_3] = 4\left[p_1^\mu p_3^\nu - g^{\mu\nu}(p_1 \cdot p_3) + p_1^\nu p_3^\mu\right]}$$

Three terms, each carrying free indices μ and ν. This is the electron's contribution.

What Are These Three Terms?

Term 1: $p_1^\mu p_3^\nu$ — first & third gamma paired, second & fourth gamma paired.
Term 2: $-g^{\mu\nu}(p_1 \cdot p_3)$ — first & second gamma paired, third & fourth gamma paired. The momenta got fully contracted into a dot product, and the free indices live on $g^{\mu\nu}$.
Term 3: $p_1^\nu p_3^\mu$ — first & fourth gamma paired, second & third gamma paired. Same momenta, but the μ and ν indices are swapped compared to Term 1.

§3.3 The "Muon" Trace — By Analogy

The second trace has the same structure with $1 \to 2$, $3 \to 4$, and indices μ, ν lowered:

"Muon" Trace (with $M = 0$)

$$\boxed{\mathrm{Tr}[\gamma_\mu \not{p}_2 \gamma_\nu \not{p}_4] = 4\left[p_{2\mu}\, p_{4\nu} - g_{\mu\nu}(p_2 \cdot p_4) + p_{2\nu}\, p_{4\mu}\right]}$$

Same three-term structure, same alternating signs.

§4 Step C: The 9-Term Contraction

What We're Doing Now

We multiply the electron trace (3 terms with upper indices μ, ν) by the muon trace (3 terms with lower indices μ, ν) and sum over μ and ν. This gives $3 \times 3 = 9$ terms. Each term is a contraction of indices — metrics eating momenta.

Write the two traces as:

$$A^{\mu\nu} \equiv p_1^\mu p_3^\nu - g^{\mu\nu}(p_1 \cdot p_3) + p_1^\nu p_3^\mu \qquad \text{(electron)}$$

$$B_{\mu\nu} \equiv p_{2\mu}\, p_{4\nu} - g_{\mu\nu}(p_2 \cdot p_4) + p_{2\nu}\, p_{4\mu} \qquad \text{(muon)}$$

We need $A^{\mu\nu} B_{\mu\nu}$ (sum over μ and ν). Let's do all 9 terms.

§4.1 The 9 Terms, One by One

Label the electron terms as A_1, A_2, A_3 and the muon terms as B_1, B_2, B_3:

$A_1 \times B_1$**:** $p_1^\mu p_3^\nu \cdot p_{2\mu} p_{4\nu} = (p_1 \cdot p_2)(p_3 \cdot p_4)$

How This Contraction Works

$p_1^\mu\, p_{2\mu}$ means: sum $p_1^0 p_{2,0} + p_1^1 p_{2,1} + \ldots = p_1 \cdot p_2$. Same for $p_3^\nu p_{4\nu} = p_3 \cdot p_4$. Two dot products multiplied.

$A_1 \times B_2$: $p_1^\mu p_3^\nu \cdot (-g_{\mu\nu})(p_2 \cdot p_4) = -(p_1^\mu p_3^\nu g_{\mu\nu})(p_2 \cdot p_4) = -(p_1 \cdot p_3)(p_2 \cdot p_4)$

$A_1 \times B_3$: $p_1^\mu p_3^\nu \cdot p_{2\nu} p_{4\mu} = (p_1 \cdot p_4)(p_3 \cdot p_2) = (p_1 \cdot p_4)(p_2 \cdot p_3)$

$A_2 \times B_1$: $(-g^{\mu\nu})(p_1 \cdot p_3) \cdot p_{2\mu} p_{4\nu} = -(p_1 \cdot p_3)(p_2 \cdot p_4)$

Why $g^{\mu\nu}\, p_{2\mu}\, p_{4\nu} = p_2 \cdot p_4$

The metric with upper indices $g^{\mu\nu}$ contracts with the lower-index momenta to form a dot product: $g^{\mu\nu} p_{2\mu} p_{4\nu} = p_2^\nu p_{4\nu} = p_2 \cdot p_4$.

$A_2 \times B_2$: $(-g^{\mu\nu})(p_1 \cdot p_3) \cdot (-g_{\mu\nu})(p_2 \cdot p_4) = (+g^{\mu\nu} g_{\mu\nu})(p_1 \cdot p_3)(p_2 \cdot p_4) = 4(p_1 \cdot p_3)(p_2 \cdot p_4)$

The Factor of 4: $g^{\mu\nu} g_{\mu\nu} = 4$

This is the trace of the 4×4 identity matrix: $g^{\mu\nu} g_{\mu\nu} = \delta^\mu_\mu = 4$. It's the dimensionality of spacetime.

$A_2 \times B_3$: $(-g^{\mu\nu})(p_1 \cdot p_3) \cdot p_{2\nu} p_{4\mu} = -(p_1 \cdot p_3)(p_2 \cdot p_4)$

$A_3 \times B_1$: $p_1^\nu p_3^\mu \cdot p_{2\mu} p_{4\nu} = (p_1 \cdot p_4)(p_3 \cdot p_2) = (p_1 \cdot p_4)(p_2 \cdot p_3)$

$A_3 \times B_2$: $p_1^\nu p_3^\mu \cdot (-g_{\mu\nu})(p_2 \cdot p_4) = -(p_1 \cdot p_3)(p_2 \cdot p_4)$

$A_3 \times B_3$: $p_1^\nu p_3^\mu \cdot p_{2\nu} p_{4\mu} = (p_1 \cdot p_2)(p_3 \cdot p_4)$

§4.2 Collecting the 9 Terms

$$
\begin{aligned}
A^{\mu\nu} B_{\mu\nu} = {} & \underbrace{(p_1 \cdot p_2)(p_3 \cdot p_4)}_{A_1 B_1} \underbrace{-(p_1 \cdot p_3)(p_2 \cdot p_4)}_{A_1 B_2} \underbrace{+(p_1 \cdot p_4)(p_2 \cdot p_3)}_{A_1 B_3} \\
& \underbrace{-(p_1 \cdot p_3)(p_2 \cdot p_4)}_{A_2 B_1} \underbrace{+4(p_1 \cdot p_3)(p_2 \cdot p_4)}_{A_2 B_2} \underbrace{-(p_1 \cdot p_3)(p_2 \cdot p_4)}_{A_2 B_3} \\
& \underbrace{+(p_1 \cdot p_4)(p_2 \cdot p_3)}_{A_3 B_1} \underbrace{-(p_1 \cdot p_3)(p_2 \cdot p_4)}_{A_3 B_2} \underbrace{+(p_1 \cdot p_2)(p_3 \cdot p_4)}_{A_3 B_3}
\end{aligned} \tag{2}
$$

§4.3 Simplifying

Group by dot-product structure:

$(p_1 \cdot p_2)(p_3 \cdot p_4)$ **terms:** $A_1 B_1 + A_3 B_3 = 2(p_1 \cdot p_2)(p_3 \cdot p_4)$

$(p_1 \cdot p_4)(p_2 \cdot p_3)$ **terms:** $A_1 B_3 + A_3 B_1 = 2(p_1 \cdot p_4)(p_2 \cdot p_3)$

$(p_1 \cdot p_3)(p_2 \cdot p_4)$ **terms:** $-1 - 1 + 4 - 1 - 1 = 0$

The Beautiful Cancellation

There are **four** terms with the structure $(p_1 \cdot p_3)(p_2 \cdot p_4)$, each with coefficient -1, and one

with coefficient +4. They sum to:

$$-1-1+4-1-1=0$$

Perfect cancellation! This always happens in the massless case. It's why the final answer is so clean.

So:

$$A^{\mu\nu}B_{\mu\nu} = 2(p_1\cdot p_2)(p_3\cdot p_4) + 2(p_1\cdot p_4)(p_2\cdot p_3)$$

The overall coefficient is $4\times 4 = 16$ from the two traces, times $g_e^4/[4(p_1-p_3)^4]$:

$$\langle|\mathcal{M}_1|^2\rangle = \frac{g_e^4}{4(p_1-p_3)^4}\cdot 16\cdot 2\big[(p_1\cdot p_2)(p_3\cdot p_4)+(p_1\cdot p_4)(p_2\cdot p_3)\big]$$

Griffiths Eq. 7.129 (Massless Case)

$$\boxed{\langle|\mathcal{M}_1|^2\rangle = \frac{8g_e^4}{(p_1-p_3)^4}\big[(p_1\cdot p_2)(p_3\cdot p_4)+(p_1\cdot p_4)(p_2\cdot p_3)\big]}$$

The coefficient is $16\times 2/4 = 8$. That's where the 8 in Griffiths comes from:
16 from the two traces (4×4), $\times 2$ from the two surviving dot-product pairs, $/\,4$ from the spin average.

§5 Converting to Mandelstam and Getting the "2"

Now rewrite the denominator for massless particles:

$$(p_1-p_3)^2 = -2(p_1\cdot p_3), \qquad (p_1-p_3)^4 = 4(p_1\cdot p_3)^2$$

So:

$$\frac{8}{(p_1-p_3)^4} = \frac{8}{4(p_1\cdot p_3)^2} = \frac{2}{(p_1\cdot p_3)^2}$$

That's where the coefficient changes from 8 to **2**.
Now translate to Mandelstam:

$$\begin{aligned}(p_1\cdot p_3)^2 &= (t/(-2))^2 = t^2/4\\ (p_1\cdot p_2)(p_3\cdot p_4) &= (s/2)(s/2) = s^2/4\\ (p_1\cdot p_4)(p_2\cdot p_3) &= (u/(-2))(u/(-2)) = u^2/4\end{aligned} \tag{3}$$

Substituting:

$$\langle|\mathcal{M}_1|^2\rangle = \frac{2}{t^2/4}\left[\frac{s^2}{4}+\frac{u^2}{4}\right]g_e^4 = \frac{8g_e^4}{t^2}\cdot\frac{s^2+u^2}{4}$$

$\langle|\mathcal{M}_1|^2\rangle$ in Mandelstam Variables

$$\boxed{\langle|\mathcal{M}_1|^2\rangle = 2g_e^4\,\frac{s^2+u^2}{t^2}}$$

$\langle|\mathcal{M}_2|^2\rangle$ follows by swapping $t\leftrightarrow u$: $\boxed{\langle|\mathcal{M}_2|^2\rangle = 2g_e^4\,(s^2+t^2)/u^2}$.

§6 The Interference Term: A Different Method

Why Griffiths Uses a Different Approach Here

For the direct terms above, Casimir's trick gave **two** traces of 4 gammas each, and we could evaluate each one separately with Theorem 13, then contract.
For the interference term $\langle \mathcal{M}_1 \mathcal{M}_2^* \rangle$, the spinors from both diagrams get tangled into one closed chain (as we saw earlier). This gives a **single** trace of **8** gamma matrices.
Applying Theorem 13 to an 8-gamma trace would require decomposing it into pairs — resulting in $7!! = 105$ terms. That's insane. So Griffiths' hint tells you to use the **contraction theorems** (Rules 8′, 9′) instead, which collapse the trace step by step.

§6.1 The Setup (from Griffiths' hint in Problem 7.37)

Following the spinor chain $3 \to 1 \to 4 \to 2 \to 3$:

$$\langle \mathcal{M}_1 \mathcal{M}_2^* \rangle = \frac{g_e^4}{4\,t\,u}\ \mathrm{Tr}[\not{p}_3\,\gamma^\mu\,\not{p}_1\,\gamma^\nu\,\not{p}_4\,\gamma_\mu\,\not{p}_2\,\gamma_\nu]$$

§6.2 Contraction #1: Contract μ Using Rule 9′

The objects between γ^μ and γ_μ are: $\not{p}_1$, γ^ν, $\not{p}_4$ (three gamma matrices).
Rule 9′: $\gamma_\mu \not{a} \not{b} \not{c} \gamma^\mu = -2\,\not{c} \not{b} \not{a}$ (reverse order, factor -2).

$$\gamma^\mu\,\not{p}_1\,\gamma^\nu\,\not{p}_4\,\gamma_\mu = -2\,\not{p}_4\,\gamma^\nu\,\not{p}_1$$

Trace becomes:

$$-2\ \mathrm{Tr}[\not{p}_3\,\not{p}_4\,\gamma^\nu\,\not{p}_1\,\not{p}_2\,\gamma_\nu]$$

§6.3 Contraction #2: Contract ν Using Rule 8′

The objects between γ^ν and γ_ν are: $\not{p}_1$, $\not{p}_2$ (two gamma matrices).
Rule 8′: $\gamma_\nu \not{a} \not{b} \gamma^\nu = 4(a \cdot b)$ (everything collapses to a number).

$$\gamma^\nu\,\not{p}_1\,\not{p}_2\,\gamma_\nu = 4(p_1 \cdot p_2) = 2s$$

Trace becomes:

$$-2 \cdot 2s \cdot \mathrm{Tr}[\not{p}_3\,\not{p}_4] = -4s \cdot 4(p_3 \cdot p_4) = -4s \cdot 2s = -8s^2$$

Interference Term

$$\boxed{\langle \mathcal{M}_1 \mathcal{M}_2^* \rangle = \frac{g_e^4}{4tu} \cdot (-8s^2) = \frac{-2g_e^4\,s^2}{tu}}$$

§7 Assembling the Final Answer

$$\langle |\mathcal{M}|^2 \rangle = 2g_e^4 \frac{s^2+u^2}{t^2} + 2g_e^4 \frac{s^2+t^2}{u^2} + \frac{4g_e^4 s^2}{tu}$$

Full Result

$$\boxed{\langle|\mathcal{M}|^2\rangle = 2g_e^4\left[\frac{s^2+u^2}{t^2}+\frac{s^2+t^2}{u^2}+\frac{4s^2}{tu}\right]}$$

Cross section (with $S = 1/2$ for identical particles, $g_e^4 = 16\pi^2\alpha^2$):

Differential Cross Section

$$\boxed{\frac{d\sigma}{dt} = \frac{\pi\alpha^2}{s^2}\left[\frac{s^2+u^2}{t^2}+\frac{s^2+t^2}{u^2}+\frac{4s^2}{tu}\right]}$$

§8 Summary: Griffiths' Two Trace Methods Compared

When to Use Which Method

Method 1: Theorem 13 + Index Contraction
Used for: $\langle|\mathcal{M}_i|^2\rangle$ (direct/exchange terms separately).
How it works: Casimir's trick gives two traces of 4 gammas each. Evaluate each with Theorem 13 (getting 3 terms with free indices), then multiply and contract. You get $3 \times 3 = 9$ terms, but most cancel.
Pros: Systematic, mechanical, hard to make sign errors.
Cons: Lots of bookkeeping (9 terms to track).

Method 2: Contraction Theorems (Rules 8′, 9′)
Used for: $\langle\mathcal{M}_i\mathcal{M}_j^*\rangle$ (interference terms).
How it works: Casimir's trick gives one trace of 8 gammas. Use contraction theorems to shrink it step by step: $8 \to 6 \to 2 \to$ number.
Pros: Fast — only 2–3 steps.
Cons: Need to know contraction theorems cold; the order of operations matters.

Bottom line: Griffiths uses both methods in the same problem. Theorem 13 when the traces are small (4 gammas). Contraction theorems when the trace is big (8 gammas).

Problem H4, following Griffiths' procedure from Examples 7.5–7.6, Eq. 7.126–7.129, and Problem 7.37.